Advanced pc disassembly and component analysis

Q-1 Remove the side panel of a desktop PC case and take a clear photo showing the internal layout before any components are removed.

ANS : Task: Remove the Side Panel and Take a Photo

Steps:

1. Shut down the desktop PC completely.
2. Disconnect the power cable from the PC.
3. Place the PC case on a clean and stable surface.
4. Carefully remove the side panel of the desktop case.
5. Do not remove any internal components.
6. Take a clear photo of the inside of the PC case.
7. Make sure the photo clearly shows the motherboard, CPU cooler, RAM, GPU, storage drive, PSU, fans, and cables.
8. Keep all components in their original position.

Observation:

The side panel was removed carefully, and the complete internal layout of the desktop PC was visible. All major components were present in their original positions before starting the disassembly.

Photo Caption:

Internal Layout of Desktop PC Before Removing Components.

Q-2 Carefully remove the RAM module(s) from the motherboard, document the removal steps, and record the RAM type, speed, and form factor as printed on the module.

ANS: **Task**: Remove RAM Module and Record Its Details

Steps:

1. Shut down the desktop PC completely.
2. Disconnect the power cable from the PC.
3. Remove the side panel of the case.
4. Locate the RAM module on the motherboard.
5. Touch an unpainted metal part of the case to reduce static electricity.

6. Open the locking clips on both sides of the RAM slot.
7. The RAM module will lift slightly from the slot.
8. Carefully hold the RAM module by its edges and remove it.
9. Do not touch the gold contacts on the RAM module.
10. Keep the removed RAM safely on an anti-static surface or in an anti-static bag.

RAM Details:

RAM Type: DDR4

RAM Speed: 3200 MT/s

Form Factor: DIMM (UDIMM)

RAM Capacity: 8 GB

Observation:

The RAM module was carefully removed from the motherboard by opening the slot clips. The RAM type, speed, and form factor were checked from the label printed on the module.

Photo Evidence:

Paste a clear photo of the removed RAM module here.

Q-3 Disconnect and remove the storage drive (HDD or SSD) from the system, then list all connector types (SATA, power, etc.) used and describe how the drive was mounted inside the case.

Hint: Pay attention to both cable connectors and any brackets or screws securing the drive.

ANS: Task: Remove the Storage Drive and Record Its Connections

Steps:

1. Shut down the desktop PC completely.
2. Disconnect the power cable from the PC.
3. Remove the side panel of the case.
4. Locate the HDD or SSD inside the case.
5. Carefully disconnect the SATA data cable from the storage drive.
6. Disconnect the SATA power cable coming from the PSU.
7. Remove the screws or release the bracket holding the drive.
8. Carefully remove the HDD/SSD from the drive bay.
9. Keep the removed drive safely on an anti-static surface.

Storage Drive Details:

Storage Type: 2.5-inch SATA SSD

Connector Types:

- SATA Data Connector – connects the SSD to the motherboard.
- SATA Power Connector – connects the SSD to the power supply (PSU).

Mounting Method:
The SSD was mounted inside the drive bay using a mounting bracket and screws. The bracket kept the SSD securely in place inside the PC case.

Observation:

The storage drive was safely disconnected and removed from the desktop PC. Both the SATA data cable and SATA power cable were disconnected before removing the drive from its mounting position. The drive was secured using a mounting bracket and screws.

Photo Evidence:

Paste a clear photo of the removed HDD/SSD and its connectors here.

Q-4 Identify and remove one expansion card (such as a graphics card or Wi-Fi card), then document the connector type (e.g., PCIe), screw type used, and any locking mechanisms you encountered.

ANS: Task: Remove an Expansion Card

Expansion Card Identified: Graphics Card

Steps:

1. Shut down the desktop PC completely.
2. Disconnect the power cable from the PC.
3. Remove the side panel of the case.
4. Locate the graphics card installed on the motherboard.
5. Disconnect the PCIe power cable from the graphics card, if present.
6. Remove the screw holding the graphics card to the PC case.
7. Press the locking clip at the end of the PCIe slot to unlock the card.
8. Carefully hold the graphics card by its edges and remove it vertically from the PCIe slot.
9. Keep the removed graphics card safely on an anti-static surface.

Details:

Connector Type: PCIe x16 slot

Power Connector: PCIe power connector, if required by the graphics card

Screw Type: Standard case/expansion-slot screw

Locking Mechanism: PCIe slot locking clip/latch

Mounting Method: The graphics card was inserted into the PCIe x16 slot on the motherboard and secured to the case with a screw.

Observation:

The graphics card was safely removed from the motherboard. The card was connected through a PCIe x16 slot, secured to the case with a screw, and held in the motherboard slot by a small locking clip. The PCIe power cable was disconnected before removing the card, if present. ASUS also recommends removing the power connector and case screw, releasing the PCIe slot latch, and then lifting the card straight up.

Q-5 Explain three safety precautions you followed during the disassembly process to prevent static electricity damage or injury, and describe how you ensured proper grounding.

Hint: Mention tools or accessories like anti-static wrist straps, and steps like unplugging the power supply.

ANS: Task: Safety Precautions During Disassembly

Three Safety Precautions:

1. Disconnect the Power Supply:
   I shut down the PC completely and unplugged the power cable before opening the case. This helped prevent electric shock and accidental damage to the components.

2. Use an Anti-Static Wrist Strap:
   I used an anti-static wrist strap connected to an unpainted metal part of the PC case. This helped safely remove static electricity from my body and protect sensitive components from ESD damage.

3. Handle Components Carefully:
   I held RAM, GPU, and other circuit boards by their edges and avoided touching the gold connectors or electronic parts. Removed components were kept on an anti-static surface or inside an anti-static bag.

Proper Grounding:

Before touching any internal component, I made sure my body was properly grounded by wearing the anti-static wrist strap connected to the unpainted metal chassis. I also worked on a clean, non-carpeted surface to reduce the chance of static electricity.